01 / PROBLEM · 00-01 MIN · FICTIONAL OFFSHORE GAS OPERATOR

Scale expert knowledge. Without guessing about safety.

NorthSea Integrity Operations depends on a few experts to investigate a growing network. Make reviewed knowledge reusable, guide newcomers and surface warnings without waiting for an AI report.

Possible release?

Corroborate physical evidence across usable sensors and failure domains.

Where to inspect?

Offer a station-supported inspection interval, never an invented exact leak coordinate.

Make expertise reusable

Explain procedures, evidence and gaps to newcomers. Knowledge transfer needs pilot validation.

Existing pressure alerts | Agent-assisted investigation | Human safety authority

02 / PROBLEM · 01-03 MIN

The prospect is buying less investigation risk.

False alarm

Unnecessary offshore personnel, vessel or ROV mobilization.

High inspection cost, operational disruption and declining trust in alarms.

Missed release

Environmental harm and liability; oil-line spills add cleanup exposure.

Serious regulatory violations can threaten licences, including possible revocation under applicable law.

> "Our dashboard is cheaper. Show that AI reduces triage effort without hiding real leaks or exporting our operational data."

Do not optimize alarm reduction while **concealing missed-event risk.**

Mobilization economics are discovery assumptions. Licence revocation is conditional, not automatic; source boundaries: engineering report.

03 / PROBLEM · 03-05 MIN

One decision. Two acceptance contracts.

Head of Subsea Operations

Budget owner: inspection effort, adoption, delivery time and accountability.

"What will four weeks prove, and who owns a wrong recommendation?"

Integrity & Instrumentation Lead

Calibration, provenance, failure domains, environmental confounders and reproducibility.

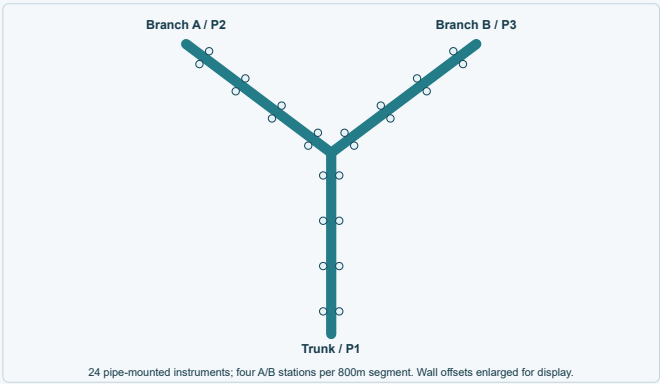
"Could a shared gateway explain these alarms? Show me the raw evidence."

My RE 2023 research motivates **accuracy, currentness, completeness and consistency.** The demo's localization is a new simplified extension, not a thesis-validated capability.

Thesis Paper C, PDF pp. 162-164; defense slides 16, 18-21 and 47. Prospect and requirements are hypotheses, not a claimed customer interview.

04 / LIVE DEMO · 05-07 MIN

Watch the window move.
Then pin the answer to evidence.



24 pipe-mounted instruments; four A/B stations per 800m segment. Wall offsets enlarged for display.

Replay → pause → inspect → ask. **60s window / 5s hop; answers stay pinned.**

Open live console Current qualification and recorded receipts

Fresh hosted investigations took about two minutes per question in the latest browser run. Label retained evidence as recorded; SCRIPTED is not LIVE.

Corroboration supports a suspected leak.

Ask the brief's questions

1. "Leak or faulty sensors?"
2. "Where should we inspect, and why?"

Open the investigation handoff: explanation, checked interval, competing causes and next review step. Click references; download the exact-window evidence packet.

S1 / seed 2026 / mature [120,180)s

Possible leak; separate sensor faults coexist.
Trunk / P1: inspect **200-600m**.
Affected stations: **300/500m**.
Unaffected brackets: **100/700m**.

Station midpoints define an inspection priority, not an estimated leak point. Verify valve/compressor changes and shared failures with the engineer.

The agent organizes the investigation. **Numerical evidence bounds the inspection area.**

Reproducible source-linked example: brief-evidence.json; current receipts: qualification.

06 / LIVE DEMO · 11-15 MIN · S2 / S4 / S5

From difficult outputs to a reviewable investigation.

S4 / Ask question three

"What evidence supports this, and what remains unknown?"

Open the valve record and before/after comparison. Read the competing causes; draft a verification question on the same window.

The operating explanation is plausible, not proof that no leak exists.

Contrast S2 / S5

S2: maintenance plus observed faults supports instrumentation review, not a proven failed component.

S5: missing observations still prevent a leak conclusion or inspection location.

An accepted abstention is **a useful, checked answer** — not a successful leak diagnosis.

Readable sources and measured comparisons precede expandable execution receipts. Recorded evidence and limitations; recorded is not a fresh live call.

Two execution loops.
One immutable evidence contract.

Source + numerical loop	Synthetic arrivals → canonical buffer → 60s/5s/5s windows → quality, hypothesis and fit ledger.
Durable boundary	PostgreSQL 16: immutable snapshots/corpus, full-text history retrieval, scoped receipts and LangGraph checkpoints.
Agent worker	Quality → diagnosis chooses search/read/compare → checked response. LangChain wraps hosted Copilot CLI inference.
Operator + telemetry	FastAPI/Pydantic → authenticated SSE → TypeScript/esbuild console. Native OTel → local Langfuse/Jaeger.

CustomerOps-style durable authority. **NumPy adds the marine numerical layer.**

08 / TECHNICAL WHITEBOARD · 19-23 MIN

Bad data stays visible.
It does not become leak evidence.

Canonicalization contract

- Known units and explicit timezones normalize; unknown formats are not guessed.
- Keep raw provenance. Flag gaps, spikes, late arrivals and duplicate conflicts.
- Conflicting values cannot silently corroborate a release.

Event-time contract

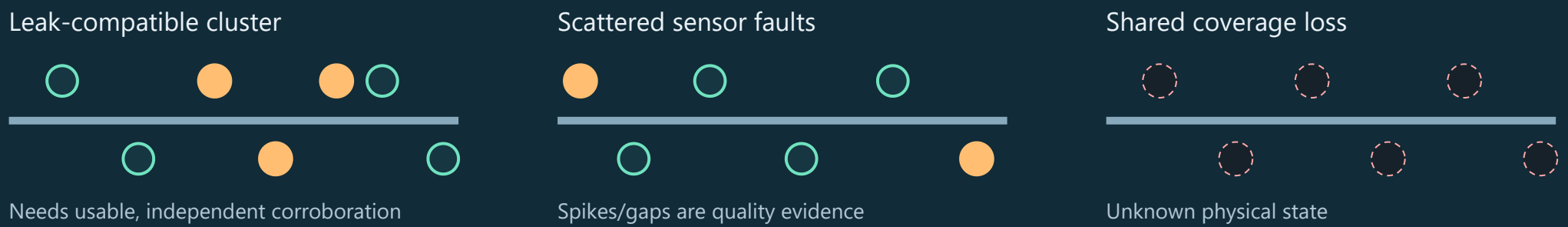
```
Window: [end - 60s, end)
Hop: 5s   Allowed lateness: 5s
Use only observations released
by the source clock.
```

Late evidence creates immutable revisions. Baseline pre-roll must be verified normal and available before the replay epoch.

History also has quality: **incident-time visibility, current approval and conflicting revisions.**

ingest.py diagnostics.py monitoring.py

Simultaneous anomalies are not sufficient.



Qualitative illustration, not copied defense geometry or a calibrated spatial model.

```
z = (value - baseline median) / scale
scale = max(1.4826 × MAD, pressure floor)
```

Frozen median/MAD stand-in for existing ML. Require overlapping negative pressure alerts, nearby connected stations and independent failure domains. Check counterevidence before suspecting a leak.

Missingness alone never proves a leak. **A shared failure can also be spatially coherent.**

A reproducible inspection candidate.
Not a confidence interval.

```
Unaffected: 100m       700m
Affected:      300m 500m
Lower = (100 + 300) / 2 = 200m
Upper = (500 + 700) / 2 = 600m
Priority interval: [200m, 600m]
400m midpoint is NOT a leak estimate.
```

Supported bracket + rejection

- Require unaffected pressure stations on both sides.
- Withhold for missing references, contradictory evidence, disconnected regions or excessive width.
- Review the interval with the integrity engineer.

Station spacing limits this heuristic. It is not a negative-pressure-wave or calibrated hydraulic localization method.

Recommend where to investigate. **Do not manufacture precision.**

localization.py: locate() synthetic.py: assumptions

Three scoped specialists.
One checked operator answer.

Sensor Quality

Checks coverage, conflicts and sensor patterns.

Quality facts are not proof of a physical event.

Event Diagnosis

Selects history searches, documents and pressure comparisons.

Cannot override numerical event/location limits.

Operator Response

Interprets engineer questions and competing causes.

Cites evidence and asks verification questions.

Observe → select evidence → act read-only → check. **Real abstention propagates.**

Code checks claims and authority. A separate same-model critic reviews every paragraph against original sources; revise once, re-review or withhold. Human review remains required.

role authority workflow grounding checks Real revision evidence

Execution must survive interruption.
The evidence must survive review.

Bounded, durable work

- LangGraph interrupt/checkpoint before external I/O; durable task/receipt transactions.
- Generation fences and idempotent intake IDs bound replay and stale workers.
- Root: at most 2 children, 10 model calls, 18 tool calls, 300 seconds.

Observable failure

- Auth/quota block is explicit; numerical monitoring continues.
- OTel spans show agent, generation, tool and guardrail in Langfuse/Jaeger.
- Stored observations are verified; exporter HTTP success alone is insufficient.

An evidence layer needs **recoverable state and truthful counters**, not just agent prompts.

Local Langfuse workflow.py telemetry.py Trace appendix

Evidence of a working demo.
Not evidence of field reliability.

120

Mature outcomes

Observable leaks: 20/20 detected/bracketed. All leaks: 20/40; S5 remains unobservable.

15/18

Agent reports

First attempts; 3 failures withheld. Explicit retries: 3/3. Fixed reports: 18/18.

129 s

Report p50

p95: 170s; fixed p95: 0.88s. Numerical warnings never wait for the LLM.

Six scenarios × three questions. All 13 operation checks pass on accepted reports. Three first-attempt failures remain in the denominator.

Accuracy and reliability first. **A faster unsupported answer still fails.**

Matched operations, cases and receipts · All cohorts and retained failures

Keep your detectors.
Buy a reviewable investigation.

Boundary	Choice	Reason / cost
Measurements and topology	Existing ML + numerical methods	Graph ML is valid; no better-coordinate claim.
Engineer questions and RCA context	Bounded read-only agent investigation	Choose evidence checks; explain competing causes.
Review and handoff	Checked, exact-window evidence packet	Readable sources; explicit gaps; human review required.

Standard reports: fixed 18/18 and faster; agents 15/18 initially, then 3/3 retries. Separate priority lab: hosted 15/16, rules 12/16. Prefer rules where equally useful.

Buy **measurable investigation value**, not a claim that only agents generalize.

Measure review effort, evidence completeness, inappropriate recommendations and total cost. No measured labor savings. Cases, caveats and receipts.

Measure triage value.
Do not sell imagined avoided spills.

Value to measure in the pilot

```
Reviewed alert volume
× verified investigation time reduction
× agreed loaded engineering cost
- integration and operating cost
```

Customer inputs are unknown. No avoided-spill value, guaranteed inspection saving or licence-loss probability is priced into this demo.

Customer numbers needed

- Assumed manual analysis: 10min. Validate this baseline.
- AI report budget: 5min. Measure human review separately.
- Expert/newcomer decision quality, total cost and integration effort.

Compare against the existing workflow. **Never trade safety sensitivity for a prettier ROI.**

A four-week shadow pilot.
One network. Explicit stop gates.

Week 1

Connector/schema, baseline, failure-domain map and security review.

Gate: approved data path and usable source contract.

Week 2

Elicit expert procedures; replay cases; adversarial grounding review.

Gate: approved knowledge and adjudicated error labels.

Week 3

Shadow workflow; expert/newcomer review; outage and load tests.

Gate: no observed unsupported/unsafe assertions.

Week 4

Accuracy, abstention, 5min report budget, knowledge transfer and cost.

Gate: quality first; revise or stop on failure.

Commit only after **scope, pricing, safety thresholds and acceptance criteria** are jointly agreed.

Conditional plan, not a guaranteed integration deadline. No sensitivity claim without labeled events or validated tests.

Keep the safety workflow.
Add a reviewable evidence layer.

What is true today

Original synthetic demo, scoped read-only tools, authenticated console, durable evidence and hosted Copilot inference.

No historian/platform connector, certified leak protection, control endpoints, production identity or security certification.

What must be approved

Customer deployment boundary, source connector, data retention and inference destination.

This demo is **not local-only**. If hosted inference is unacceptable, resolve the architecture before real data is used.

> "Will you sponsor a scoped shadow pilot, subject to integrity and security approval, with agreed stop gates and a joint baseline review?"

Expert knowledge becomes **reviewable criteria and evidence** — not unchecked model authority.

18 / OUT OF ROLE · 55-60 MIN

Questions for AITOMATIC.

1. How do your strongest FDEs turn an expert's operating judgment into governed agent behavior — and who owns ongoing validation?
2. Where do customer pilots most often stall: data access, integration, evidence quality or workflow adoption?
3. What would you expect me to own from discovery through production handoff in my first 90 days?

5m	10m	30m	10m	5m
Problem 00-05	Demo 05-15	Whiteboard 15-45	Business 45-55	Questions 55-60

Private repository Customer brief Technical appendix

APPENDIX A / ON DEMAND · NO EXTRA SCHEDULED TIME

Test the counterexamples. Not just the easy leak.

Scenario	Challenge	What the reviewer should inspect
S1	Nearby pressure drops plus a separate fault	Independent support, inspection bracket; history does not prove cause.
S2	Scattered spikes, gaps and drift	Current maintenance versus stale revision; matched sensor comparisons.
S3	Full-source telemetry outage	Source clock advances; physical state remains unknown.
S4	Broad pressure change	Current valve log plus measured change; concurrent leak remains possible.
S5	Leak with too few observers	Telemetry work order cannot restore missing physical evidence.
S6	Normal pressure variability	No qualifying event is not a safety certification.

History qualification covers S1/S2/S4/S5; the numerical hold-out covers all six. Synthetic context labels are not field root-cause labels.

APPENDIX B / ON DEMAND

Can you prove the agents actually ran?

Receipts, not role labels

- LIVE mode, requested/observed gpt-6.1-sol, identity source cli_event.
- Task/generation/window scope, model/tool counters and durable accepted result.
- Reuse linked explicitly; not counted as a new specialist execution.

Stored observability

Open the pressure cohort's **verified_trace_ids** in Langfuse.

Agent, generation, tool and guardrail observations must be stored, not just HTTP-accepted. Fresh browser feeds do not recreate historical events.

Current topology, exact question and pinned evidence. **No substituted historical cohort.**

Current LIVE receipts and boundaries Brief example evidence

APPENDIX C / ON DEMAND

Failures changed the implementation. They did not change the truth standard.

Observed failure	Root-cause correction	Evidence boundary
Empty hosted history searches	Ranked keyword matching; contextual acceptance beyond numerical answers.	Initial 6/12 contexts retained, not advertised as a pass.
Follow-up input overflow	Compact input, one schema copy, finite 48k-character envelope; topology-expanded asset filters.	Call/time and physical-certainty gates unchanged.
Missing checkpoint bootstrap	Explicit migration, fail-fast startup, fresh/reopen/handoff tests.	Integrated database proof beyond isolated unit tests.
Long ingest lock / UI race	Bounded idempotent chunks; Agents focus before async task read.	Targeted contention and held-read browser regression.

Senior FDE ownership means **reproducing, fixing and retaining evidence**, not hiding demo failures.

APPENDIX D / ON DEMAND

Implemented boundaries. Unfinished production work.

Implemented in the demo

- Argon2 login, scoped cookies/CSRF and authenticated API/SSE.
- Forced tenant/site PostgreSQL RLS; separate tool/checkpoint roles.
- Static read-only authority; inference subprocess strips app/database/viewer credentials.
- Loopback-bound local services and secrets outside the repository.

Not certified or integrated

- Production SSO/multi-user identity, HA and security certification.
- Historian, Falkor or DanaOS connector.
- Field-calibrated gas/oil detection or localization.
- Local-only inference or autonomous shutdown/control.

Research grounding: thesis Paper C (RE 2023), defense 16/18-21/47. Public industry positioning: Falkor. Regulatory scenario: archived Norwegian Petroleum Act translation, sections 10-13/10-16; not current legal advice.

Original deck and diagrams. CustomerOps informs customer-first narrative, scoped specialists, durable evidence, failure ownership and explicit tradeoffs; its content/assets and private discussions are not copied. Detailed source boundaries: REPORT.md.